Extra credit problems

Math 427

0. Find a mistake or misprint in the book. (The score depends on the type of mistake.)

1. Describe all the motions of the Manhattan plane.

2. Construct a metric space $\mathcal{X}$ and a distance-preserving map $f\colon \mathcal{X} \to \mathcal{X}$ that is not a motion of $\mathcal{X}$.

3. Note that the following quantity

$$\tilde{\measuredangle} ABC = \left[\begin{array}{lll} \pi & \text{if} & \measuredangle ABC = \pi, \\ -\measuredangle ABC & \text{if} & \measuredangle ABC < \pi. \end{array}\right.$$

can serve as the angle measure; that is, the axioms hold if one changes everywhere $\measuredangle$ to $\tilde{\measuredangle}$.

Show that $\measuredangle$ and $\tilde{\measuredangle}$ are the only possible angle measures on the plane, but without Axiom IIIc, this is not longer true.

4. Show that a composition of three reflections in the sides of a nondegenerate triangle does not have a fixed point.

5. Lines ℓ and m are tangent to two circles of radiuses r and R on such a way the circles are on one side of ℓ and on different sides of m. Let A and B be tangential points of ℓ and Q be the point of intersection ℓ and m. Show that

$$QA{\cdot}QB = R{\cdot}r.$$

6. Given a line segment with a marked midpoint, make a ruler-only construction a line thru a given point P parallel to the segment, and prove that the construction works. (You may play with applet www.geogebra.org/m/nq8y9xhq.)

7. Let ABC be a nondegenerate triangele and $A' \in (BC)$, $B' \in (CA)$, $C' \in$ $\in (AB)$ be the points such that

$$2{\cdot}\angle AA'B \equiv 2{\cdot}\angle BB'C \equiv 2{\cdot}\angle CC'A \equiv \tfrac{2}{3}{\cdot}\pi.$$

Show that the triangle formed by the lines (BB'), (CC'), and (AA'), is congruent to $\triangle ABC$.